

myElectron Solar IoT

Open-Source ESP32 Solar-Powered IoT Development Platform

Integrated solar charging and **18650 battery** power management

Hardware-based **ATtiny13A** power cycling for longevity

LoRaWAN and **4G-LTE** modular expansion

IP65 enclosure readiness for outdoor use



Hardware Features, Interfaces & Applications

A complete solar-powered IoT platform for robust deployment and versatile development

BUILT-IN HARDWARE

ESP32-WROOM 32 MCU
CN3065 solar charger
& 5V boost converter
ATTiny13A hard power
cycle controller
Tilt switch, status LED,
reset/program switches,
waterproof rocker switch

EXPANSION INTERFACES

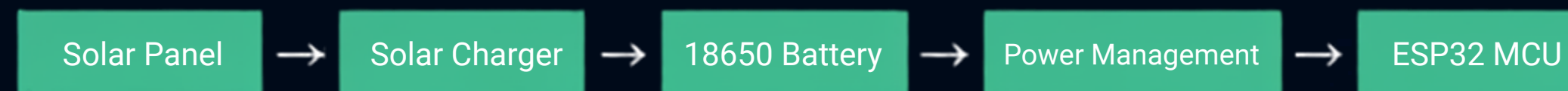
GPIO & sensor header
SCL, SDA, i034, i035,
i014, i027, 3.3V, 5V, GND

LoRaWAN & 4G-LTE
modular headers for
flexible connectivity

ECOSYSTEM COMPATIBILITY

myElectron solar panels
18650 batteries
LoRaWAN modules
Sensor probes
IP65 enclosures

POWER FLOW



REAL-WORLD APPLICATIONS

Smart agriculture
Environmental & air
quality monitoring

Remote telemetry &
weather stations

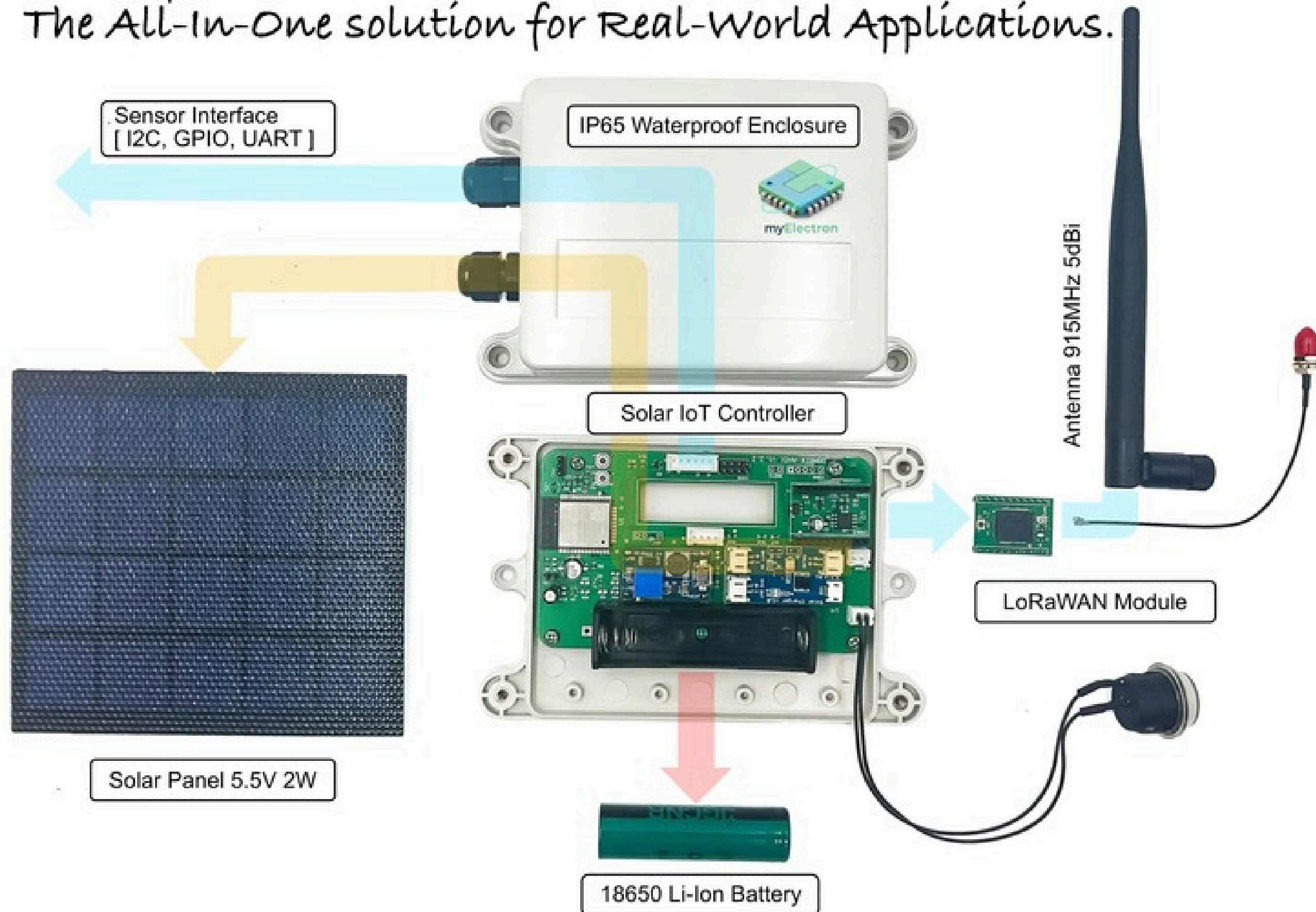
Off-grid LoRaWAN/
cellular sensor nodes

Embedded systems
development &
prototyping



SOLAR IoT : ESP32 DEVELOPMENT KIT

Build Your Own Solar-Powered IoT Node.
The All-In-One solution for Real-World Applications.





UART Interface
for debug/Program/
Serial Device Interface

ESP32 WROOM-32
Wi-Fi/BLE+4Mb Flash

Integrated TILT Switch
apply for wireless OTA/
Tampered Proof function

5V DC-DC Step-Up
Boost Converter

18650 Rechargeable
Li-Ion Battery Holder

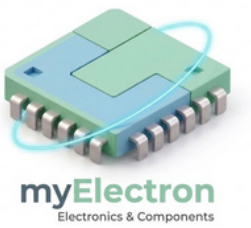
I2C / GPIO / 3.3V / 5V
for Sensor Interface.

LoRa / 4G-LTE
Expansion Interface

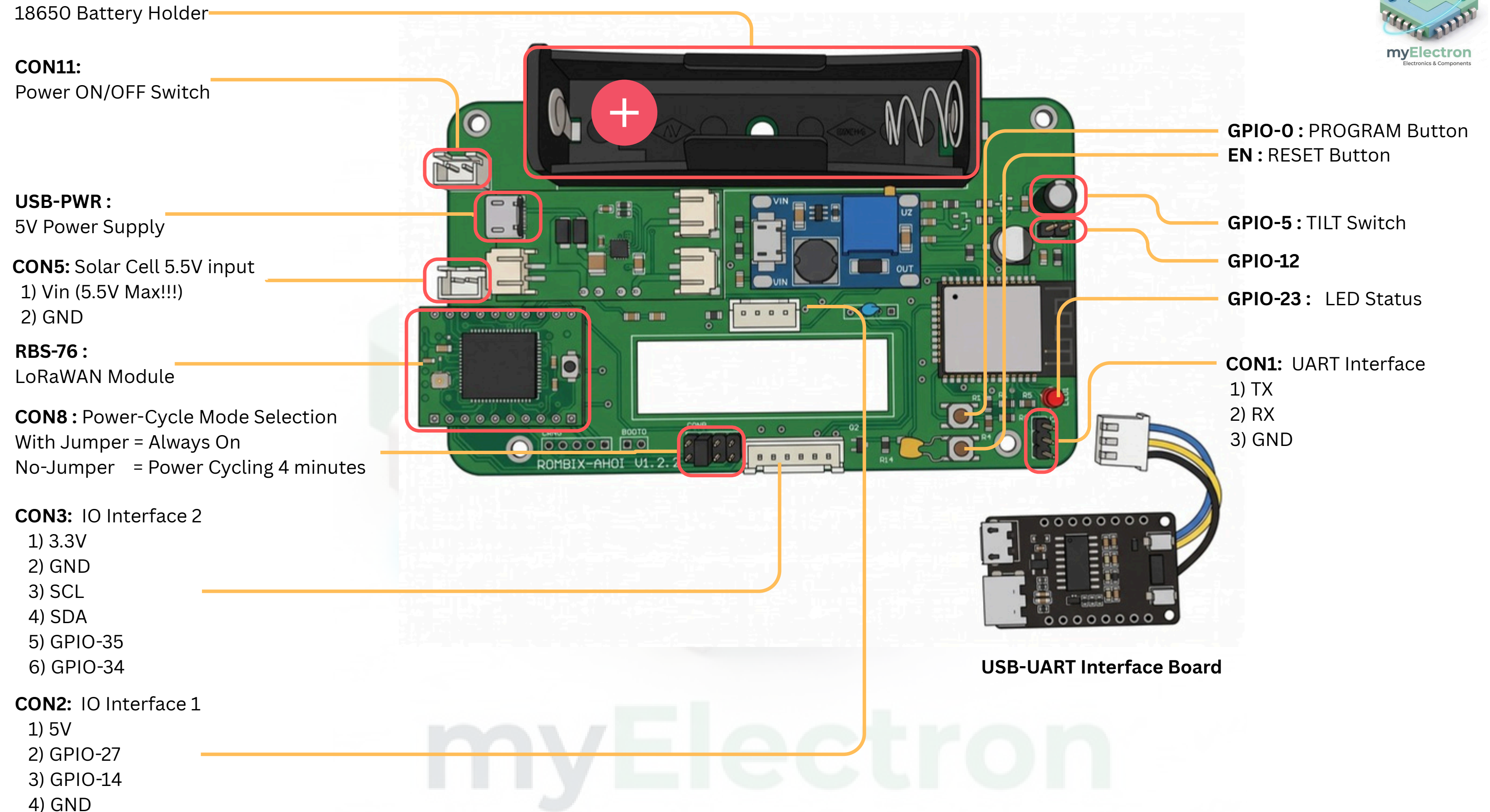
ATTINY13A : Supervisory
Power Management
Automatic Power Cycling

Integrated Solar Charger
Adaptive Charging Control

Waterproof
Power On/Off Switch

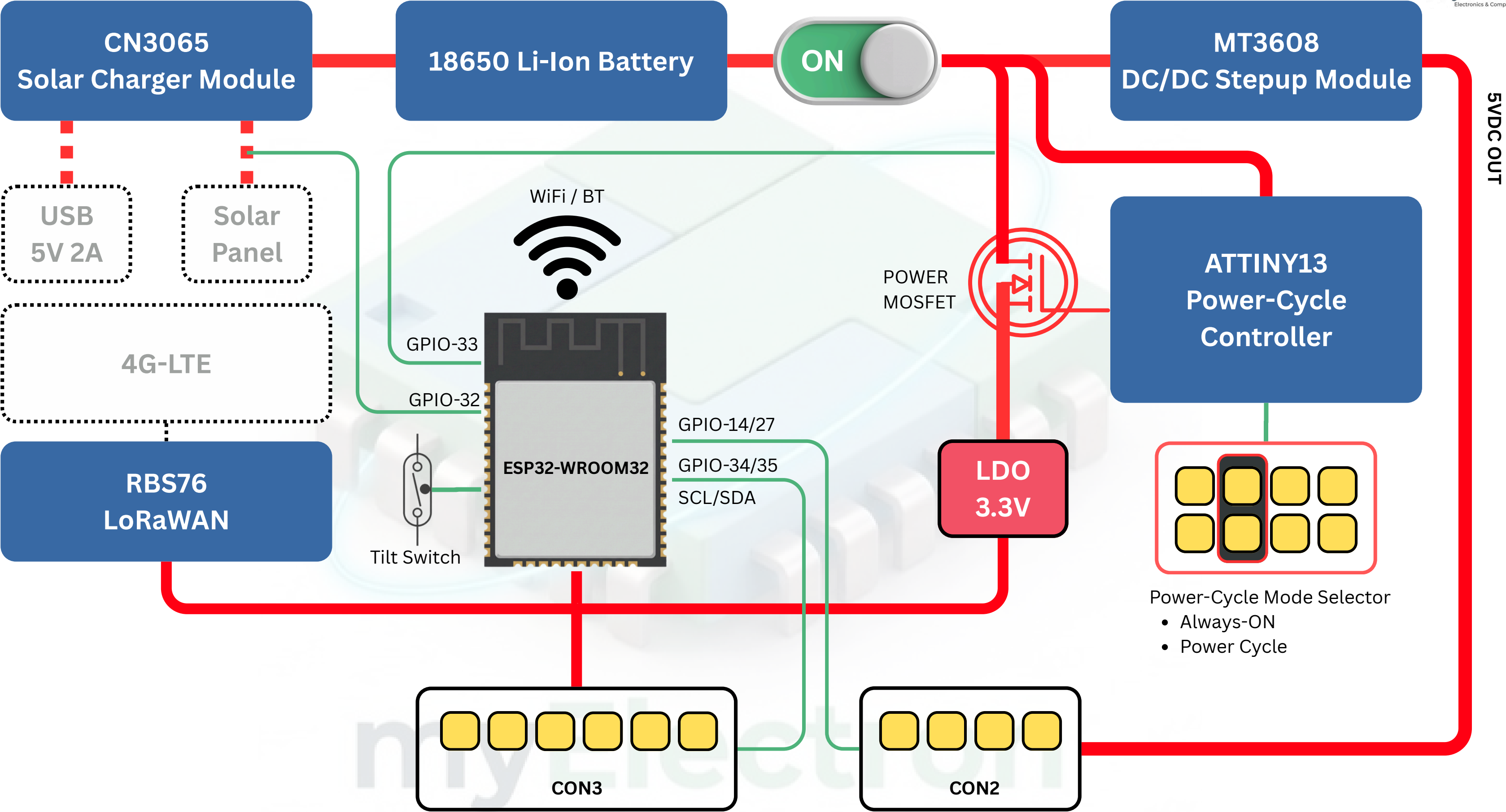
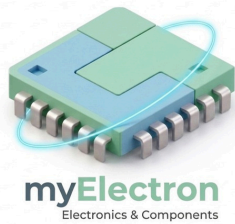


myElectron : Solar IoT Controller

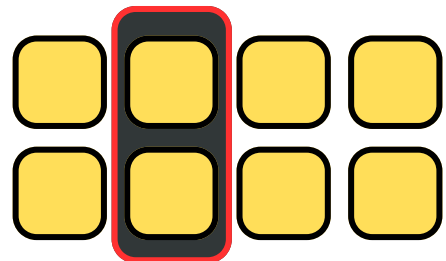
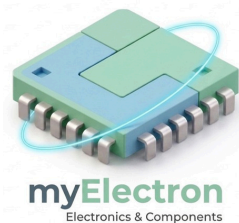


myElectron Solar-IoT Controller Pinout Diagram

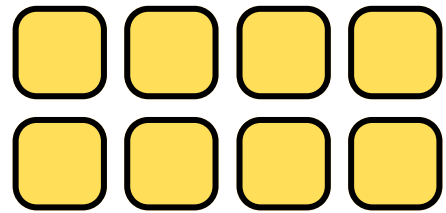
myElectron Solar-IoT Controller Block Diagram



myElectron Solar-IoT Controller GPIO Pinout Diagram



ALWAYS-ON Power



Power-Cycle

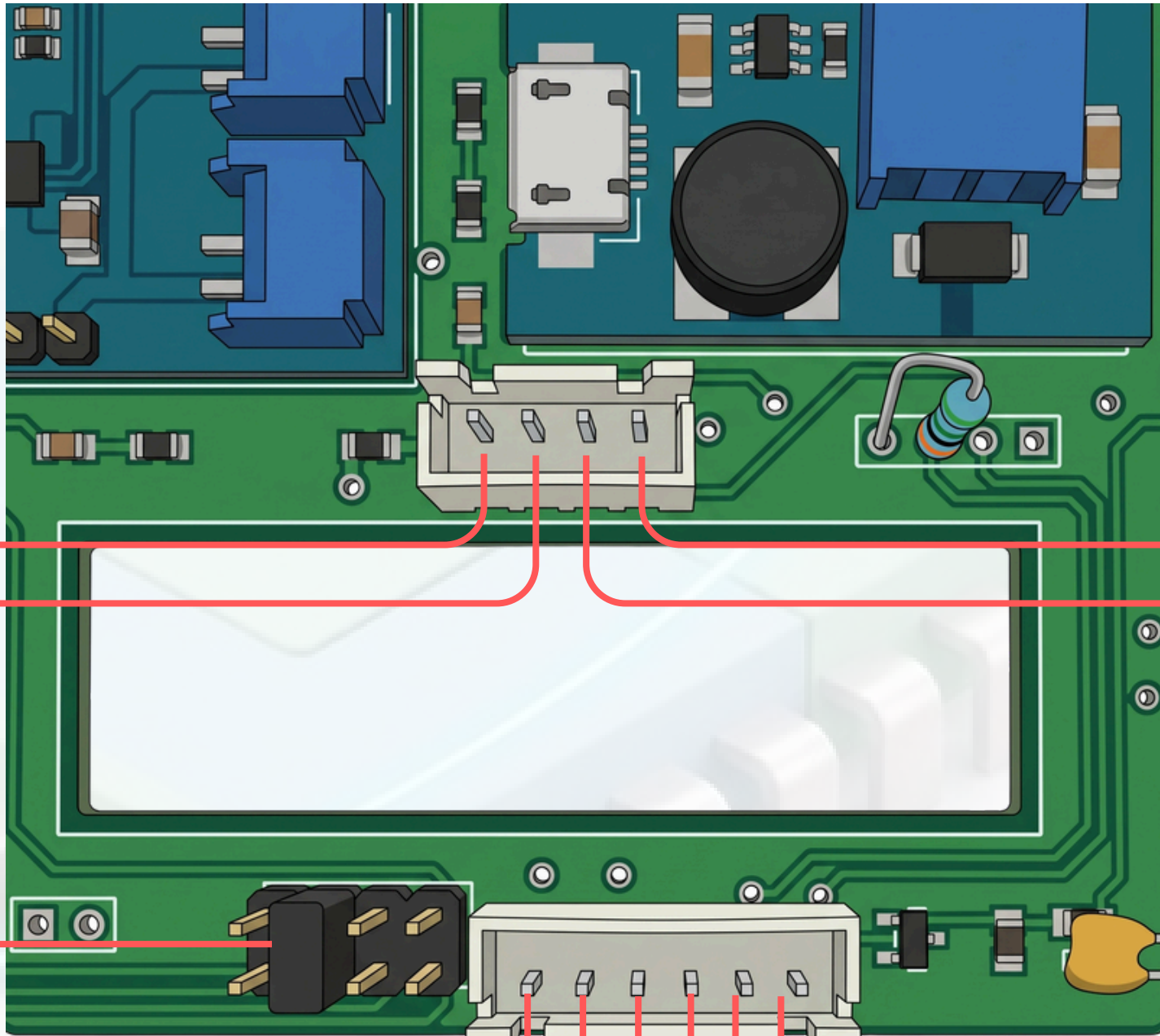
- ON :1 minute
- OFF : 4 minutes

GND
GPIO-14

5VDC OUT
GPIO-27

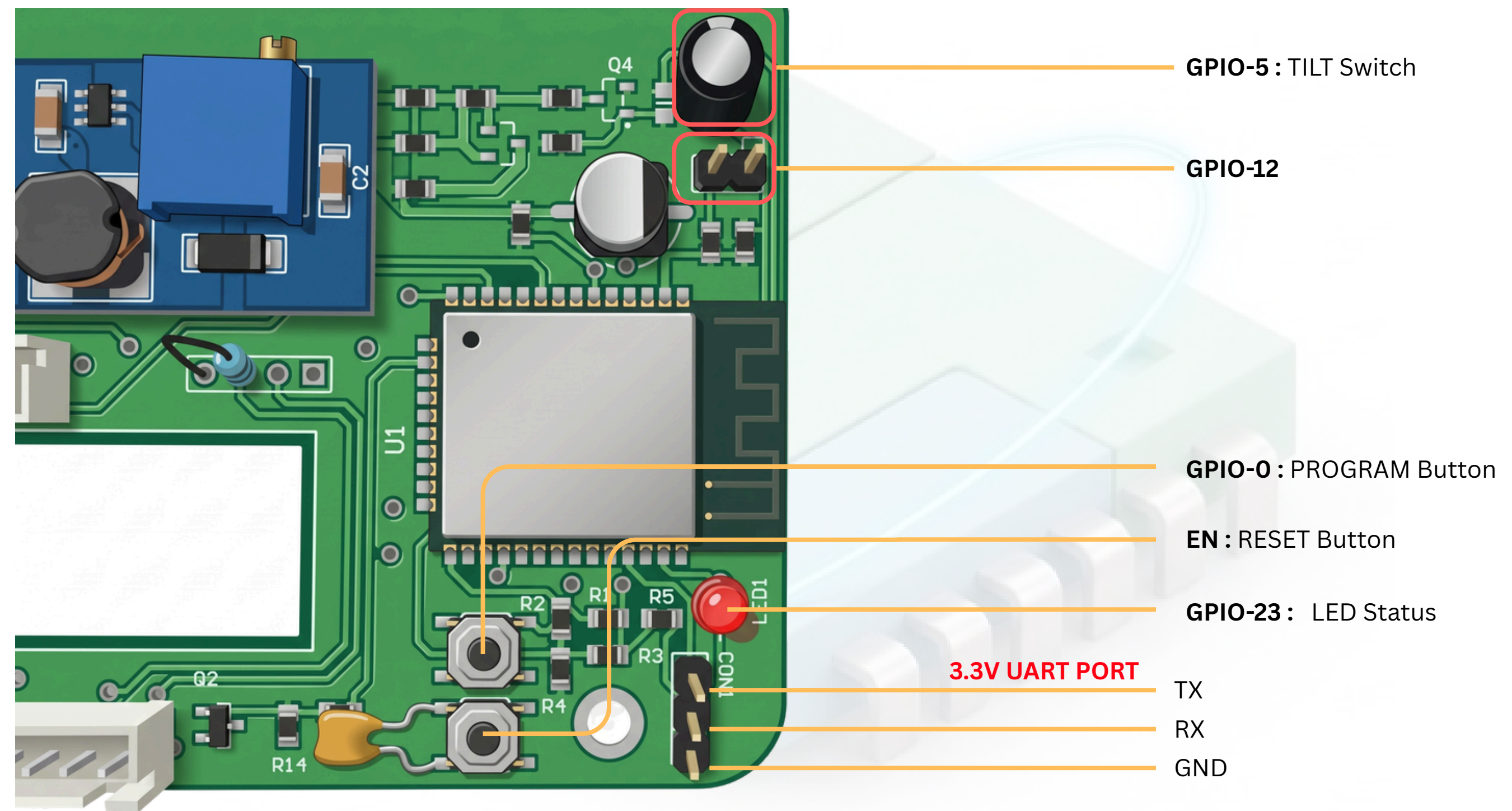
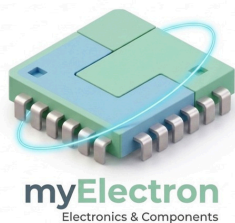
3.3V
GND
SCL

GPIO-34
GPIO--35
SDA



myElectron

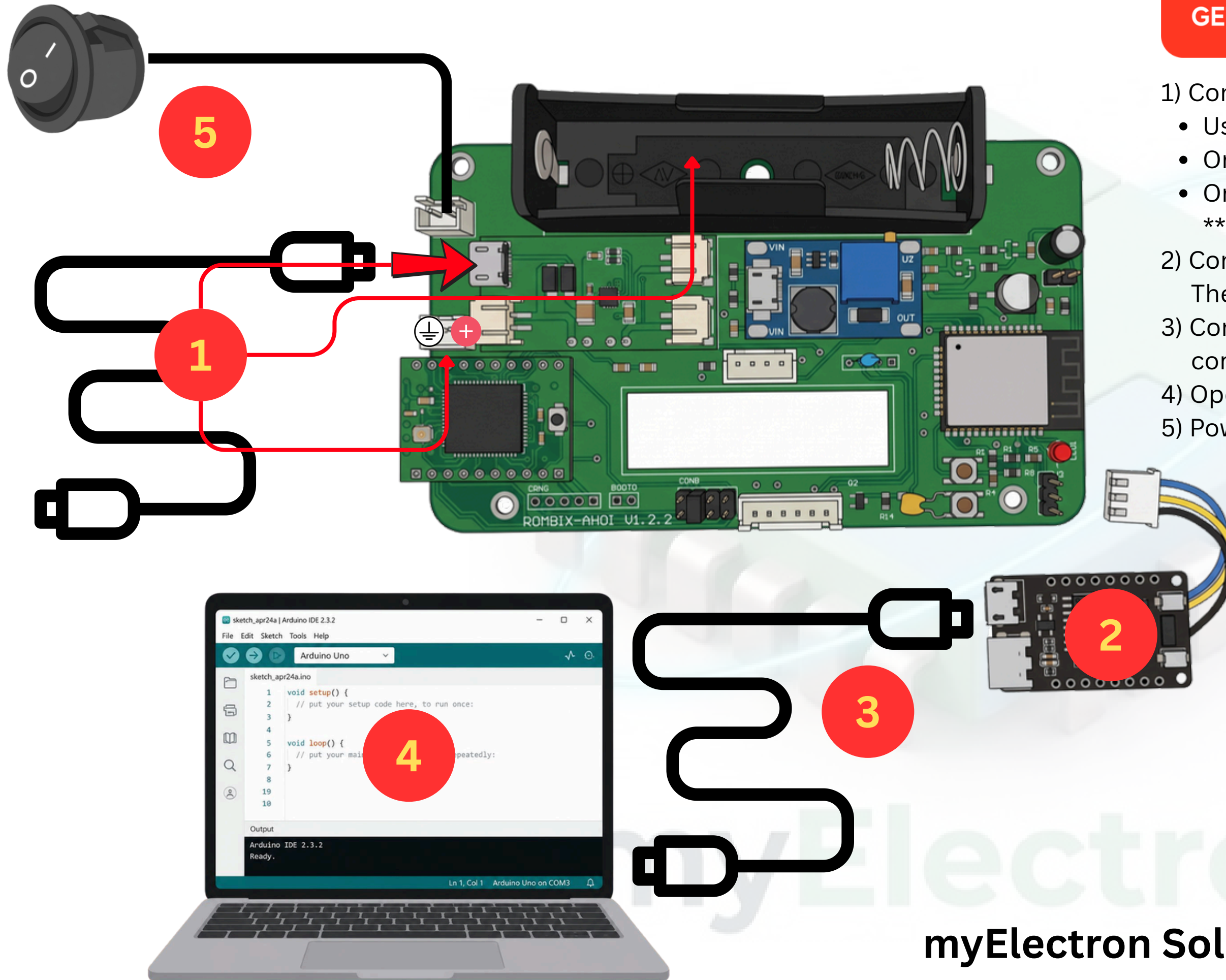
myElectron Solar-IoT Controller ESP32 PG/RES/UART Pinout Diagram



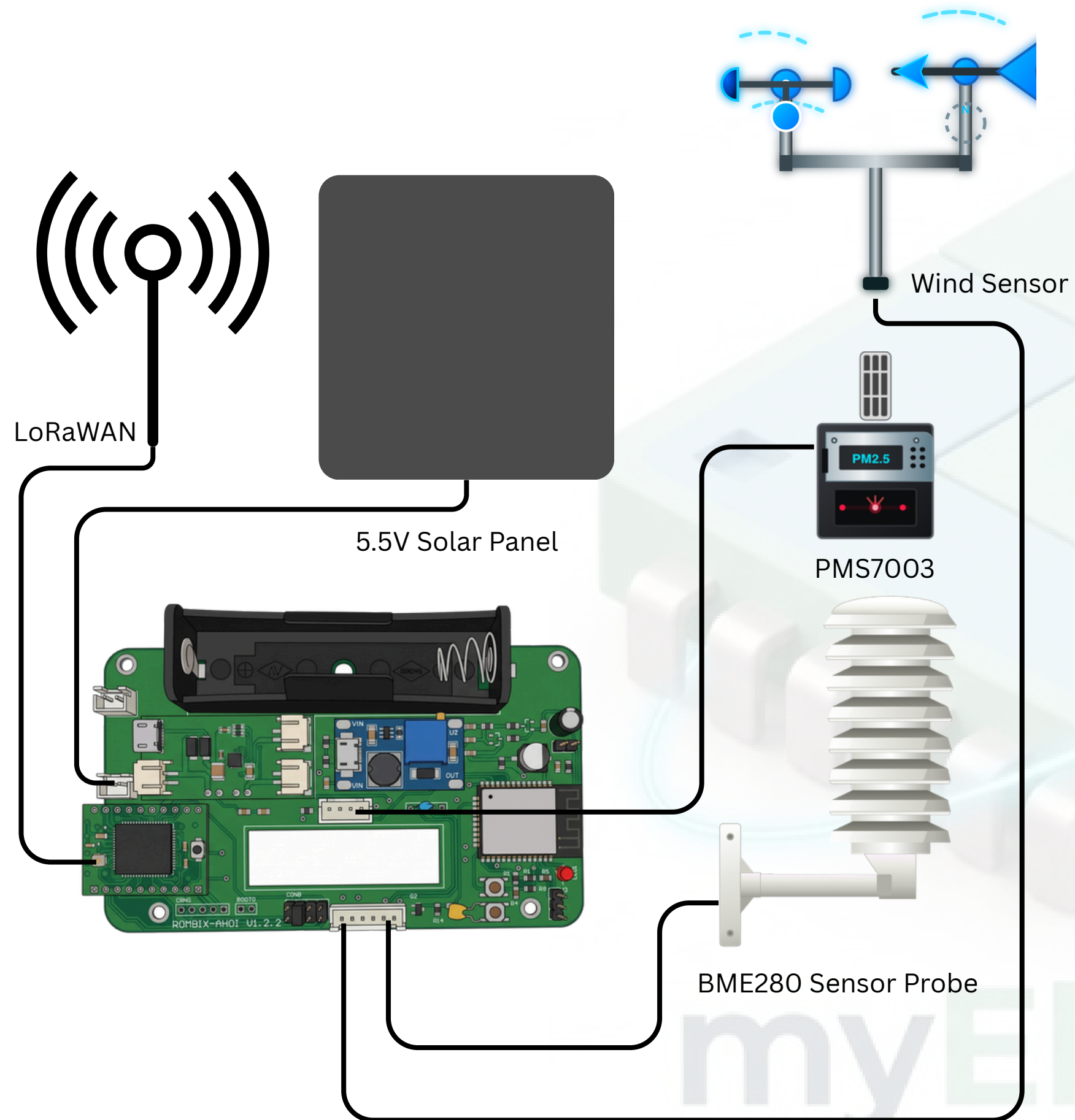
GET STARTED →

- 1) Connect 5V Power Supply,
 - Using USB power cable. (recommended 5V 2A)
 - Or insert battery Li-Ion 18650 into batt. holder.
 - Or Supply 5VDC on CON11 (Solar Cell power-port)

*** Caution : 5.5V Max. voltage ***
- 2) Connect USB-UART Interface board.
The black cable connect ground.
- 3) Connect USB Data cable to your PC and set your serial com port @115200/8/N/1
- 4) Open your serial com.port
- 5) Power on the Solar-IoT board.



myElectron Solar-IoT Controller Getting Start



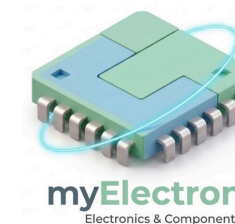
What is Factory Self-Test

The Solar-IoT Controller will be preloaded at the factory with a system test program. User can record the serial number and then overwrite the existing program. The following will be tested:

- Read the chip ID (used as the board's serial number)
- Onboard battery voltage ADC
- Onboard solar PV voltage ADC
- Tilt switch status
- I2C (connected to the BME280 sensor probe)
- Pulse counter (no value, the sensor not connected)
- PMS7003 laser dust sensor (no value, the sensor not connected)
- Test communication with the LoRaWAN module



What is showing on Factory Self-Test Screen



AQI-AH01 V1.101.17

ChipID : XXXXXXXXXX *** This is your Serial No. Please noted. ***

LOOP :1

LoRa SLEEP >> sleep 900 sec uart_on

SETTING Vref...

VB = 0

OK

ADC 1425

BATT 3.66 V

Scantime = 22000

H1-max = 65535, H1-min = 65535

H2-max = 65535, H2-min = 65535

ID : XXXXXXXXXX

SN : XXXXXXXXXX

TX : 7

Init DTHP... OK!

*** System initial for factory self-test***

Vref 1895

ADC 1425

BATT 4.35 V

TILT: 0

ADC 12

PV 0.04 V

T: 30.48 °C H: 66.22 %RH P: 100876.98 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.74 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.58 Pa

AN_H1_Max: 65535 AN_H1_Min: 65535 AN_H1_value: 0

AN_H2_Max: 65535 AN_H2_Min: 65535 AN_H2_value: 2245

Pulse : 1 RPM : 0.00 Speed : 0.00 m/s

Read-Time : [5219]

Using xPM01= 0

Using xPM25= 0

Using xPM10= 0

Using Last PM1 = 0 PM2.5 = 0 PM10 = 0

*** Initial I2C for BME280 sensor probe ***

*** Checking battery ADC on board ***

*** Checking Tilt Switch Status ***

*** Checking Solar PV voltage ***

*** Checking BME280 ***

*** Checking Pulse counter ***

- Not connected here

*** Checking PMS7003 ***

- Not connected here

MSG = 4b01b300040be719d199ecea000.....

Initial LoRaWAN...

DEBUG-LOG >> Ok

SET ADR-OFF >> Ok

SET DR1 >> Ok

>> Ok

SET TX-CNF >> Ok

MAX-EIRP 7

>> Ok

>> Ok

DevEUI: 990990FEAA194806>> Ok

DevADDR: AA194806>> Ok

>> Ok

Join >> accepted

Sending...

....

....

....

....

LoRa SLEEP >> sleep 900 sec uart_on

>> err

Sending 3 time in 19s.

On-Time = 34 sec.

SLEEP 400 sec.

*** Checking RBS76-LoRaWAN Module ***

- Only signal test, no gateway connected.

*** Enter Deep-Sleep ***